

## Topic 6 – Microbiology, Immunity and Forensics

Candidates will be assessed on their ability to:

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| <b>6.1</b>  | understand the principles and techniques involved in culturing microorganisms, using aseptic technique                                                                                                                                                                                                                     |
| <b>6.2</b>  | understand the different methods of measuring the growth of microorganisms, as illustrated by cell counts, dilution plating, mass and optical methods (turbidity)                                                                                                                                                          |
| <b>6.3</b>  | understand the different phases of a bacterial growth curve (lag phase, exponential phase, stationary phase and death phase) and be able to calculate exponential growth rate constants                                                                                                                                    |
| <b>6.4</b>  | <b>CORE PRACTIAL 13</b><br><b>Investigate the rate of growth of microorganisms in a liquid culture, taking into account the safe and ethical use of organisms.</b>                                                                                                                                                         |
| <b>6.5</b>  | (i) be able to compare the structure of bacteria and viruses (nucleic acid, capsid structure and envelope) with reference to Ebola virus, tobacco mosaic virus (TMV), human immunodeficiency virus (HIV) and lambda phage ( $\lambda$ phage)<br>(ii) understand what is meant by the terms <i>lytic</i> and <i>latency</i> |
| <b>6.6</b>  | understand how <i>Mycobacterium tuberculosis</i> and human immunodeficiency virus (HIV) infect human cells, causing symptoms that may result in death                                                                                                                                                                      |
| <b>6.7</b>  | (i) know the major routes pathogens may take when entering the body<br>(ii) understand the role of barriers in protecting the body from infection, including skin, stomach acid, and gut and skin flora                                                                                                                    |
| <b>6.8</b>  | understand the non-specific responses of the body to infection, including inflammation, lysozyme action, interferon and phagocytosis                                                                                                                                                                                       |
| <b>6.9</b>  | understand the roles of antigens and antibodies in the body's immune response including the involvement of plasma cells, macrophages and antigen-presenting cells                                                                                                                                                          |
| <b>6.10</b> | understand the differences between the roles of B cells (B memory and B effector cells), and T cells (T helper, T killer and T memory cells) in the host's immune response                                                                                                                                                 |
| <b>6.11</b> | understand how individuals may develop immunity (natural, artificial, active and passive)                                                                                                                                                                                                                                  |
| <b>6.12</b> | understand how the theory of an 'evolutionary race' between pathogens and their hosts is supported by evasion mechanisms shown by pathogens                                                                                                                                                                                |
| <b>6.13</b> | understand the difference between bacteriostatic and bactericidal antibiotics                                                                                                                                                                                                                                              |
| <b>6.14</b> | <b>CORE PRACTICAL 14</b><br><b>Investigate the effect of different antibiotics on bacteria.</b>                                                                                                                                                                                                                            |
| <b>6.15</b> | know how an understanding of the contributory causes of hospital-acquired infections has led to codes of practice regarding antibiotic prescription and hospital practice that relate to infection prevention and control                                                                                                  |

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| <b>6.16</b> | know the role of microorganisms in the decomposition of organic matter and the recycling of carbon                                                                                              |
| <b>6.17</b> | know how DNA can be amplified using the polymerase chain reaction (PCR)                                                                                                                         |
| <b>6.18</b> | know how gel electrophoresis can be used to separate DNA fragments of different length                                                                                                          |
| <b>6.19</b> | understand how DNA profiling is used for identification and determining genetic relationships between organisms (plants and animals)                                                            |
| <b>6.20</b> | understand how to determine the time of death of a mammal by examining the extent of decomposition, stage of succession, forensic entomology, body temperature and degree of muscle contraction |

# Unit 5: Respiration, Internal Environment, Coordination and Gene Technology

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## IA2 compulsory unit

## Externally assessed

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### Unit description

#### Introduction

Following on from energy and the environment in Unit 4, this unit starts by considering energy within organisms and how energy is made available for processes, including muscle contraction. There are further details of some of the topics in AS, including coordination of the heartbeat. Students will also consider some aspects of maintenance of the internal environment, with specific references to kidney function and the mode of action of hormones. This leads on to the topic of coordination in mammals and in flowering plants, the effects of drugs on the nervous system and how modern techniques of gene technology are used for the production of drugs. Modern aspects of gene technology, including the use of microarrays, provide a foundation for further study in this area.

#### Practical skills

In order to develop their practical skills, students should be encouraged to carry out a range of core practical experiments related to this topic. There are also opportunities to carry out additional practical work, such as investigating habituation.

#### Mathematical skills

There are opportunities for the development of mathematical skills in this unit, including using ratios, constructing and interpreting frequency tables, bar charts and histograms and the use of statistical tests. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

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## Assessment information

- First assessment: June 2020.
  - The assessment is 1 hour and 45 minutes.
  - The assessment is out of 90 marks.
  - Candidates must answer all questions.
  - The paper may include multiple-choice, short-open, open-response, calculations and extended-writing questions.
  - The pre-released scientific article will underpin one question of the paper. This question carries 20 marks.
  - The paper will include synoptic questions that may draw on two or more different topics.
  - The paper will include a minimum of 9 marks that target mathematics at Level 2 or above.
  - Calculators may be used in the examination. Please see *Appendix 8: Use of calculators*.
  - Candidates will be expected to apply their knowledge and understanding to familiar and unfamiliar contexts.
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